Competitive Analysis:
Socio-Psi vs. MicroPsi2

Alignment, Gaps, Opportunities, and Collaborative Leverage

Socio-Psi Project

Technical Report — February 2026

Abstract

This document provides a systematic comparative analysis of the Socio-Psi and MicroPsi2, evaluating alignment with Psi theory, identifying architectural gaps, and proposing opportunities for extension and potential collaboration. We assess each subsystem across functional dimensions and provide a strategic roadmap for incorporating the most impactful elements of Psi theory into the Socio-Psi's existing architecture.

Contents

1 Executive Summary

The Socio-Psi and MicroPsi2 share a common ancestor in Dörner's Psi theory (Dörner & Güss, 2013) but diverge significantly in implementation philosophy:

- MicroPsi2 (Bach, 2012) faithfully implements the full Psi architecture via node nets, spreading activation, and formal modulator dynamics. It is a *cognitive modeling framework*—general-purpose, theoretically pure, and environment-agnostic.
- Socio-Psi takes Psi drives as a foundation and extends them with Jungian archetypal dialogue, LLM-powered cognition, genuine hardware embodiment, and semantic memory. It is a *situated application*—specific, expressive, and running on real hardware.

Neither system subsumes the other. MicroPsi2 has architectural depth (modulators, node nets, ReCoN scripts) where Socio-Psi has experiential richness (archetypal voices, weighted memory, speech recognition, creative output). The most productive path forward is selective adoption of Psi mechanisms into the Socio-Psi, not convergence.

2 Alignment Analysis

2.1 Drive System

The Socio-Psi's drive system is directly inspired by and closely aligned with Psi theory.

Table 1: Drive-by-drive alignment between Socio-Psi and Psi theory

Drive	Psi	S-P	Notes
Energy	✓	✓	Both: metabolic / battery level
Integrity	✓	✓	Both: structural safety
Arousal	✓	✓	Both: optimal stimulation
Competence	✓	✓	Both: mastery, success/failure ratio
Certainty	✓	✓	Both: predictability, resolved uncertainty
Affiliation	✓	✓	Both: social connection
Curiosity	∼	✓	Psi: implicit in competence/certainty; Jung: explicit drive
Recognition	∼	✓	Psi: within social drives; Jung: explicit drive
Individuation	—	✓	Jungian extension: growth toward wholeness

Alignment Score: Drives

85% aligned. Core Psi drives are present and function correctly. Socio-Psi extends with curiosity (reasonable), recognition (reasonable), and individuation (Jungian contribution not in Psi). Drive dynamics (rise rates, fall rates, demand clamping to baseline) follow Psi principles.

2.2 Drive Dynamics

Both systems implement homeostatic demand dynamics:

Table 2: Drive dynamics comparison

Mechanism	MicroPsi2	Socio-Psi
Demand signal	Deviation from setpoint	`demand` rises toward 1.0, clamped at `baseline`
Satisfaction	Via node net activation reaching goal	Via `SATISFACTION_MAP`: action results feed back
Pleasure/pain	Drive demand decrease/increase	`delta` field: negative = pleasure, positive = pain
Urgency	Demand + anticipation weighting	`demand * (1 + max(0, delta * 10))`

2.3 Somatic Grounding

Strongest Alignment

Both projects emphasize embodiment, but the Socio-Psi is *more* concretely embodied than MicroPsi2. Where MicroPsi2 uses abstract world adapters, the Socio-Psi maps drives directly to real hardware sensors: battery level, CPU load, thermal state, RAM pressure, fan speed, lid position, network connectivity, ambient light, keyboard touch, and Bluetooth device proximity.

This is genuine somatic grounding—the agent's drives are satisfied or frustrated by the *actual physical state* of its body, not by simulated sensor inputs.

3 Gap Analysis

3.1 Gap 1: Modulator Layer

Severity: High. The modulator layer is the single most impactful missing component. Without it, all drives produce identical cognitive processing regardless of their urgency or configuration. A panicking agent and a curious agent both "think" the same way—only their action selections differ.

In Psi theory, modulators shape the *quality* of thought:

- High arousal (many urgent drives): broad, shallow scanning; faster action selection; coarser discrimination. The agent "panics"—it looks at everything but sees nothing deeply.
- Low arousal (drives satisfied): narrow, deep processing; slower but more careful action selection. The agent "contemplates."
- Low selection threshold (recent failures): impulsive behavior; any modest signal triggers action.
- High securing rate (high competence): thorough outcome verification; the agent double-checks results.

Current state: The Socio-Psi has no modulator layer. The `heartbeat` interval varies with somatic state (stressed → faster cycles), which is a crude proxy for arousal modulation but doesn't affect cognitive processing depth.

Proposed implementation: See Section 6.

3.2 Gap 2: Emergent Emotions

> Severity: Medium. The agent tracks drive states and produces "feeling" labels (`Pain: curiosity rising`), but these are descriptive labels, not emergent configurations that alter behavior. True Psi emotions would arise from modulator states and influence action selection, archetype activation, and speech content.

Current state: The `[FEELING]` section of drive perception lists drives that are rising (pain) or falling (pleasure). The archetypal dialogue system passes drive states as context to all four archetype prompts, influencing their *tone* and *concerns* rather than gating which archetypes speak. Infrastructure for drive-weighted archetype activation exists (`calculate_weights` in `dialogue.py`) but is not yet integrated into the main loop—all four voices currently speak every cycle. This is a partial implementation: drive states shape archetype *content* but not archetype *activation strength*.

3.3 Gap 3: Node Nets / Spreading Activation

> Severity: Low (for current architecture). MicroPsi2's node nets provide learned, associative action selection. The Socio-Psi uses a static `SATISFACTION_MAP` and LLM-based action selection instead. For a system that relies on an LLM as its cognitive engine, learned node nets would be redundant—the LLM already provides flexible, context-dependent action selection.

Current state: Action selection is hybrid: (1) `DRIVE_SUGGESTIONS` maps urgent drives to candidate actions, (2) the LLM proposes actions based on full perceptual context, (3) high-urgency drives "prime" actions directly. This works well for the current architecture.

Trade-off: Node nets would allow the agent to *learn* action associations over time rather than using a static map. However, the LLM already adapts based on conversation history, and the memory weight system provides a form of experiential learning. The cost-benefit of implementing node nets is low.

3.4 Gap 4: Hierarchical Planning (ReCoN)

> Severity: Medium-Low. The Socio-Psi selects flat action lists (1–3 actions per cycle). It cannot decompose "search the web for something interesting" into sub-steps. However, the LLM implicitly handles sequencing—in practice, the agent's action selections are contextually appropriate without explicit hierarchical planning.

3.5 Gap 5: Expectation Horizon

> Severity: Medium. The agent has no forward model. It does not anticipate future drive states or plan actions based on expected outcomes. Each cycle is reactive: sense → think → act. Adding even a simple expectation horizon ("if I don't charge soon, energy will reach critical in N cycles") would produce more interesting anticipatory behavior.

4 Competitive Advantages

The Socio-Psi has several capabilities that MicroPsi2 lacks entirely:

4.1 Archetypal Internal Dialogue

Shadow

"Three hours dark. Did they leave forever?"

Anima

"I keep listening for footsteps."

Persona

"Stay ready. They might return."

The Socio-Psi generates *multiple simultaneous internal voices* via parallel LLM calls to archetype-specific prompts. The Ego mediates these into a coherent thought. This produces psychologically rich inner monologue that MicroPsi2—with its purely numerical activation dynamics—cannot generate.

Competitive edge: The archetypal dialogue system makes the agent's inner life *legible*. You can read the Shadow's fear, the Anima's longing, and the Persona's composure. This is not just richer output—it's a fundamentally different kind of cognitive transparency.

4.2 Genuine Hardware Embodiment

MicroPsi2 connects to environments through abstract world adapters. The Jung Agent runs *on the body it inhabits*. Its drives are satisfied or frustrated by the actual state of its MacBook: real battery levels, real thermal readings, real CPU load, real network connectivity.

Competitive edge: No simulation gap. The agent's somatic experience is not a model of a body—it *is* the body's state.

4.3 LLM-Powered Cognition

MicroPsi2 uses node nets for all cognition. The Socio-Psi uses a multi-model LLM architecture via Ollama: Cydonia 22B (custom `jung-mid` model) for archetypal voices and action selection, phi4 for creative generation (poetry, dreams, observations, piano compositions), and optionally Anthropic Claude Haiku for the psyche query. This enables:

- Natural language inner monologue (not numerical activations)
- Context-sensitive action selection that adapts to novel situations
- Creative output: poetry, observations, dream narratives, piano compositions
- Speech: the agent *speaks aloud* via macOS TTS
- Hearing: continuous speech recognition via `SFSpeechRecognizer`

4.4 Semantic Memory with Weighted Recall

The Socio-Psi's memory system uses sentence-transformer embeddings for semantic search (cosine similarity over 384-dimensional vectors) and a weight-based recall system where meditation boosts surfaced memories' weight by +0.3 (capped at 5.0) while decaying non-surfaced memories by ×0.95. Sampling uses `weight × current_strength`, where strength follows intensity-weighted exponential decay. This differs from MicroPsi2's protocol chain mechanism, where action-outcome chains strengthen through successful drive satisfaction and decay through disuse.

4.5 Meta-Cognition

The `MetaCognition` module tracks the agent's own thoughts, analyzes harmony trends, and generates LLM-powered reflections on its psychological state. This higher-order monitoring has no equivalent in MicroPsi2.

5 Collaborative Leverage

5.1 What Socio-Psi Should Adopt from Psi/MicroPsi2

Highest-value adoption: The modulator layer (Bach, 2012b). A small, pure-arithmetic module that would make cognitive processing *qualitatively different* under different drive configurations—the single most impactful Psi mechanism missing from the Socio-Psi.

1. Modulator layer (high priority): Derive arousal, resolution, selection threshold, and securing rate from current drive states (Bach, 2009). Use these to modulate LLM parameters (temperature, top-p), cycle timing, archetype weighting, and action verification behavior.
2. Drive-weighted archetype activation (high priority): Wire the existing `calculate_weights` method in `dialogue.py` into the main loop. This is low-hanging fruit—the infrastructure exists but is not yet called.
3. Emergent emotion labels (medium priority): Map modulator configurations to emotion names (Bach, 2012b). Display current emotion in the console and feed it into the LLM context. This makes the agent's emotional state legible without requiring discrete emotion tracking.
4. Expectation horizon (medium priority): Maintain a simple forward model of drive trajectories. Use it to generate anticipatory actions ("battery trending toward critical—conserve now") and to enrich the LLM context with predicted future states.
5. Satisfaction-weighted learning (low priority): Track which actions actually satisfy which drives over time (rather than using a static `SATISFACTION_MAP`). Adjust mappings based on experience. This would make the agent's behavior adaptive.

5.2 What MicroPsi2 Could Learn from Socio-Psi

1. Archetypal multi-voice cognition: Internal dialogue between competing psychological perspectives produces richer, more nuanced behavior than single-voice node net activation.
2. LLM integration: Using language models as the cognitive substrate enables natural language thought, creative output, and adaptive behavior without hand-coding node nets.
3. Hardware embodiment: Running directly on hardware sensors eliminates the simulation gap and produces genuine somatic grounding.

4. Individuation as a drive: The concept of a long-term growth drive toward psychological integration is a Jungian contribution absent from Psi theory.

6 Strategic Roadmap

6.1 Phase 1: Modulator Layer (High Impact)

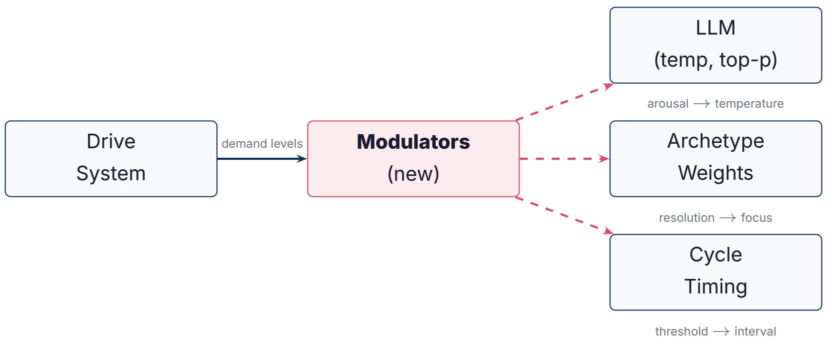

Figure 1: Proposed modulator layer: drive states derive modulators, which shape LLM parameters, archetype weighting, and cycle timing.

Implementation sketch for `modulators.py`:

- `arousal` = mean urgency across all drives. Maps to LLM temperature (high arousal → higher temp → more divergent, "panicked" thinking).
- `resolution` = 1 − arousal, weighted by certainty satisfaction. Maps to top-p (high resolution → lower top-p → more focused generation).
- `selection_threshold` = exponential moving average of recent action success rate. Maps to minimum urgency required to prime actions.
- `securing_rate` = competence satisfaction level. Maps to whether the agent re-checks action results in subsequent cycles.

6.2 Phase 2: Emergent Emotions

Define emotion as a named region in modulator space:

```
EMOTION_MAP = {
    "anger":     {"arousal": (0.7, 1.0), "resolution": (0.0, 0.3),
                  "competence_frustrated": True},
    "fear":      {"arousal": (0.7, 1.0), "integrity_threatened": True},
    "joy":       {"arousal": (0.0, 0.4), "resolution": (0.6, 1.0),
                  "recent_satisfaction": True},
    "curiosity": {"arousal": (0.3, 0.6), "resolution": (0.6, 1.0),
                  "certainty_low": True},
    "sadness":   {"arousal": (0.0, 0.3), "affiliation_deprived": True},
}
```

Display current emotion in the cycle output and feed it into the LLM context as `[EMOTION] anger (high arousal + competence frustration)`.

6.3 Phase 3: Expectation Horizon

Maintain a 3–5 cycle forward projection of drive trajectories based on current rise/fall rates. Generate anticipatory actions when projected states cross urgency thresholds. Feed projections into the LLM context:

```
[FORECAST] energy will reach critical in  4 cycles at current drain
```

7 Risk Assessment

Table 3: Risk assessment for Psi theory integration

Risk	Likelihood	Impact	Mitigation
Modulator layer adds latency per cycle	Medium	Low	Modulators are pure arithmetic—nanosecond cost
Emotion labels feel "canned"	Medium	Medium	Use as LLM context input, not console display; let the LLM interpret
Over-engineering beyond observable benefit	High	Medium	Implement Phase 1 first, evaluate, then decide on Phase 2–3
Divergence from Psi orthodoxy	Certain	None	Socio-Psi is not a Psi implementation—it's a Psi-*inspired* Jungian system

8 Architectural Comparison Diagram

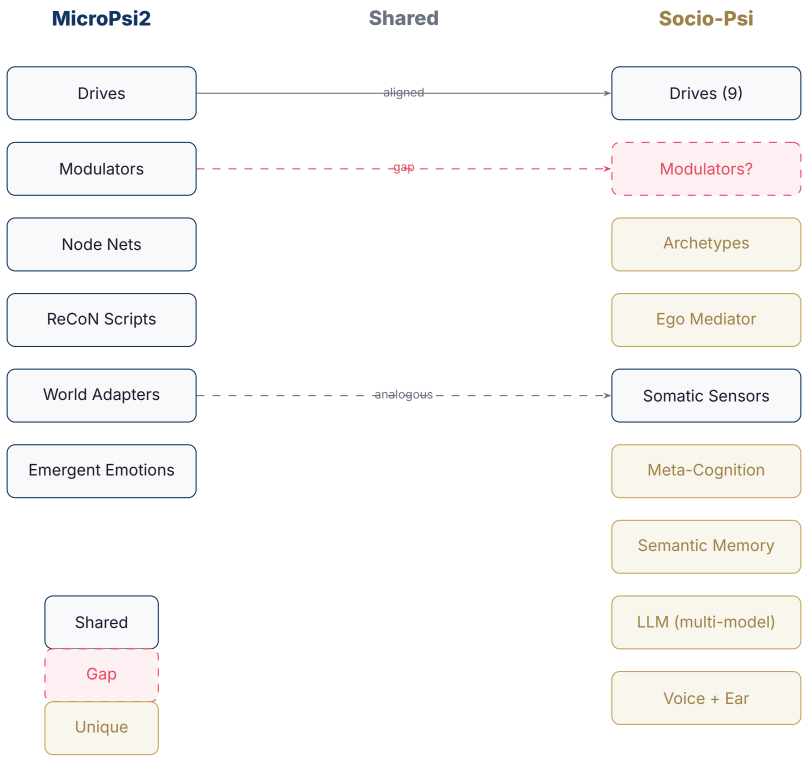

Figure 2: Architectural comparison: shared components (drives), gaps (modulators), and unique capabilities (archetypes, LLM, voice). Dashed red = missing in Socio-Psi. Gold = unique to Socio-Psi.

9 Conclusion

The Socio-Psi is not a MicroPsi2 clone and should not become one. Its strength lies in the synthesis of Psi-theoretic drives with Jungian archetypal dynamics, LLM-powered cognition, and genuine hardware embodiment—a combination no other system offers.

The highest-value adoption from MicroPsi2 is the modulator layer: a small, elegant mechanism that would make the agent's cognitive processing *qualitatively different* under different drive configurations. This single addition would bring the Socio-Psi closer to Psi theory's most powerful insight—that motivation doesn't just select *what* you do, it shapes *how* you think.

Everything else—node nets, ReCoN scripts, formal spreading activation—can be left to MicroPsi2. The Socio-Psi's LLM-based cognition already provides the flexibility that these mechanisms were designed to achieve, through a fundamentally different (and arguably more expressive) computational substrate.

The path forward is not convergence but *complementarity*: let MicroPsi2 be the rigorous cognitive model, and let Socio-Psi be the living, breathing, speaking silicon psyche that demonstrates what these ideas feel like from the inside.

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